



THE SOCIETY / APARTMENT MANAGEMENT PORTAL

Team ID: 7 | Section: S10 | Members: V. Sujit - 2520090085 | D. Bala Charan - 2520090087 | M. Sai Chaitanya - 2520090023

Guide: Dr. Divya Priya

Database Systems Engineering and Distributed Backend Development (25C51302E)
KLH Bachupaly Campus, CSE | PBL 2026-27

Abstract

- A centralized web portal for apartment administrators and residents to manage resident/flat records, maintenance billing and payments, complaints, visitor gate Passes, amenity bookings and communication.
- Stack: React.js, ReactAPI (Python) and MongoDB with Motor/PyMongo/Beanie.
- Outcome: organized, convenient and centralized society operations.

Tools & Technologies

- Front-End: React.js, UI or HTML/CSS/JavaScript
- Back-End: FastAPI (Python) + REST APIs
- Database: MongoDB
- Driver / ODM: Motor (Async) or PyMongo / Beanie
- Tools: Postman, Swagger UI, Git, VS Code

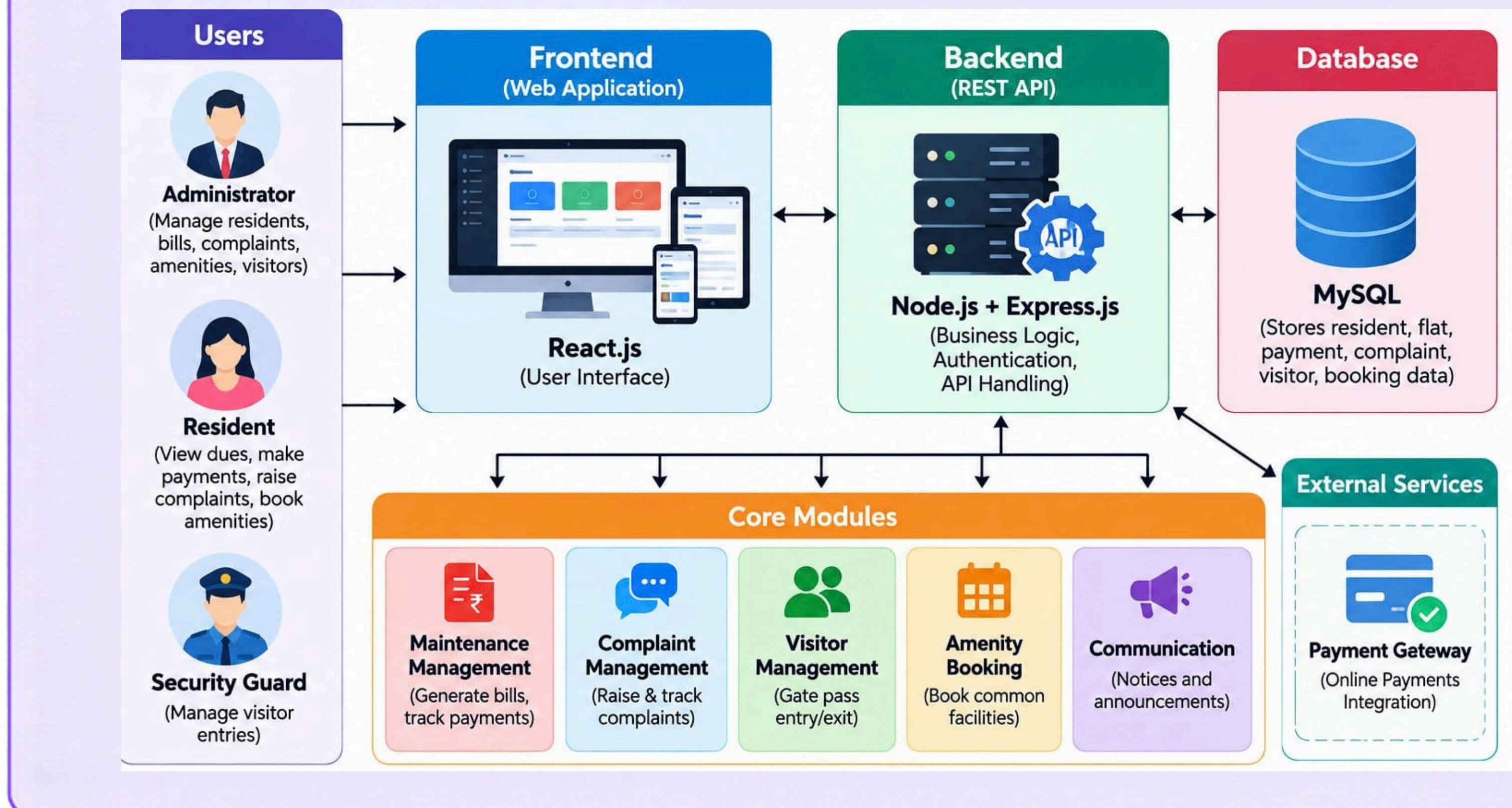
Problem Statement & Objectives

- Problem: Manual or separate systems can cause inefficient records, delayed communication, and difficulty tracking payments and requests.
- Objective: 1: Centralize resident, flat and society records
- Objective 2: Digitize billing, payments, complaints and visitor entry
- Objective 3: Support amenity booking and resident communication
- Scope: Residents + administrators; records, billing, payments, complaints, visitors and bookings.

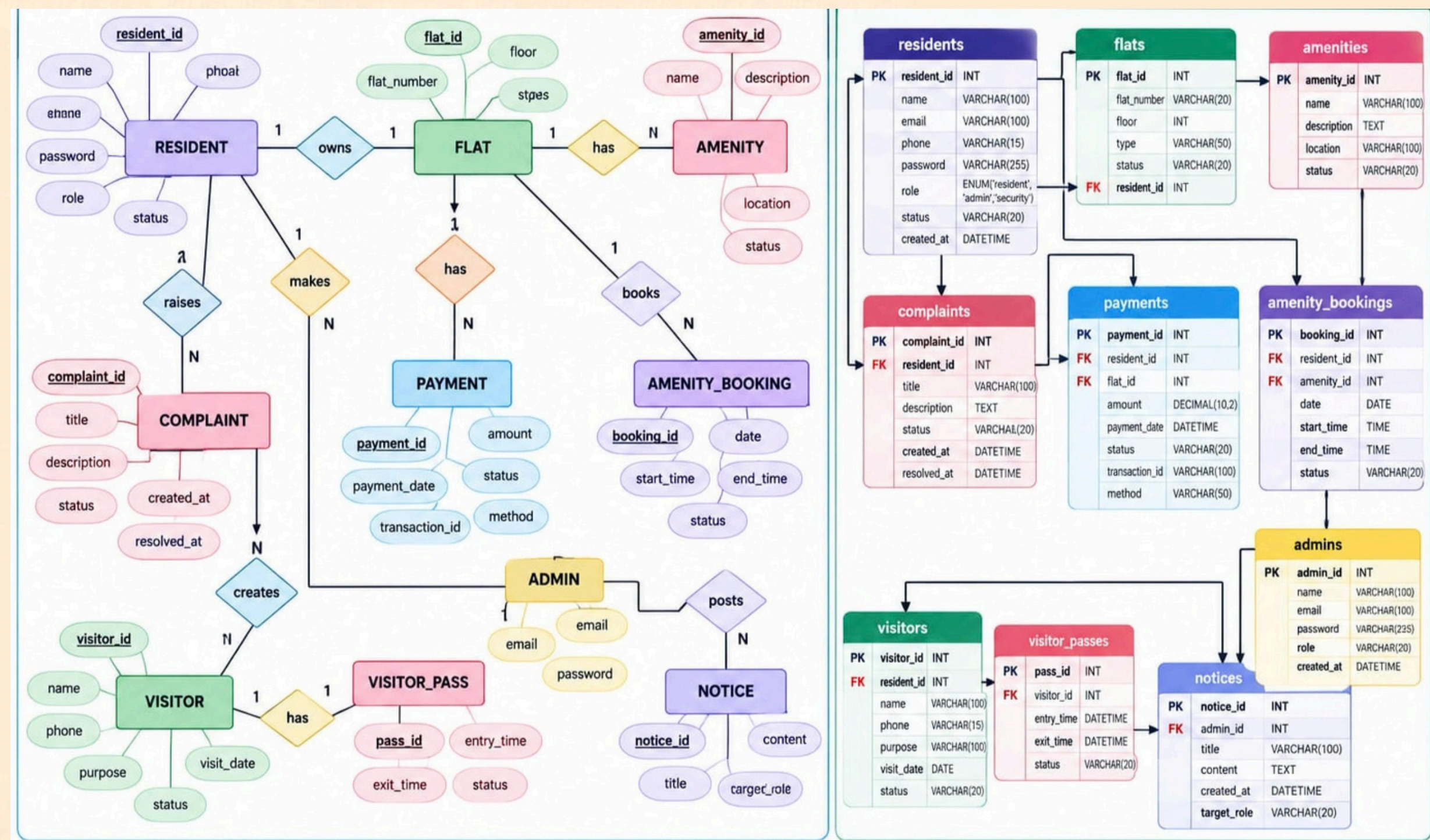
Existing vs Proposed System

Problem	Existing System	Proposed System
Records	Manual /legards	Centralized portal
Maintenance	Difficult to track	Generate a track bills
Visitors	Manual entry	Gate pass management
Regainst.	Scattered tracking	Complains & bookings
Linstation / Gap	Fragmented workflows	One centralized platform

System Architecture



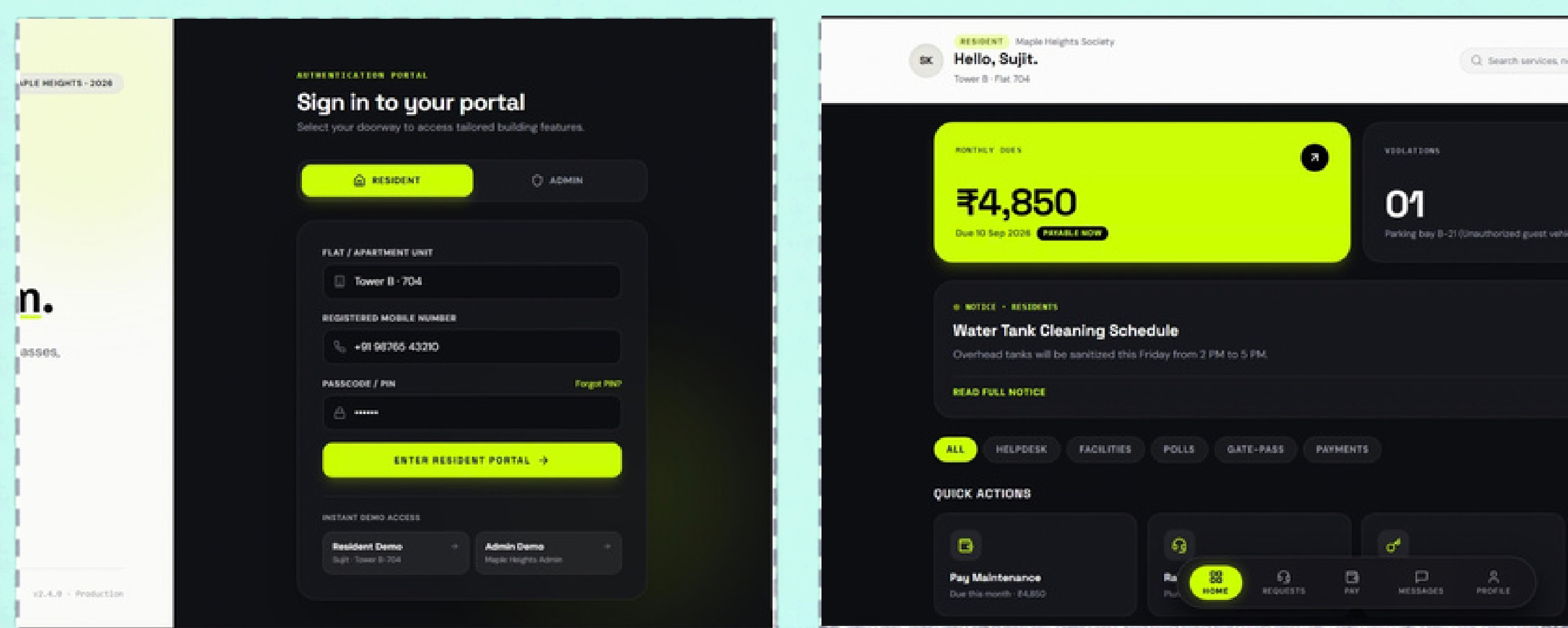
ER Diagram & Database Design



Feasibility Study

- Technical: React.js / HTML-CSS J5, FastAPI, MongoDB and APItesting tools are specified.
- Economic: Uses an open-source-oriented software stack.
- Operational: Centralizes routine society activities for administrators and residents.
- Schedule: Designed for completion within the trimester project timeline.

Implementation Screenshots



Results & Testing

- Validation focus:
- CRUD operations for core records
- API testing with Postman / Swagger UI
- Tracking payments and requests
- Consistent storage of society data

Conclusion

- The portal brings key apartment-management activities into one web-based platform, supporting targeted records, online maintenance payments, visitor management and dwddent services.

Future Scope

- Extend resident communication workflows
- Expand amenity and visitor management feature
- Add further service services as requirements evolve